

09e — Normalization

Key Concepts

Term	Definition
Normalization	Organizing a DB to reduce redundancy by splitting data into related tables
Denormalization	Intentionally duplicating data to avoid expensive joins — for performance
Redundancy	Same data stored in multiple places — leads to inconsistency
Data Integrity	Data stays accurate and consistent across the database
Transitive Dependency	Non-key column depends on another non-key column (violates 3NF)
Partial Dependency	Non-key column depends on part of a composite key (violates 2NF)

Normal Forms

Normal Form	Rule	Fixes
1NF	Atomic values — no lists or repeating groups in a cell	Repeating groups
2NF	No partial dependency on composite key	Column depends on part of key
3NF	No transitive dependency between non-key columns	Non-key depends on non-key

Examples

1NF Violation:

OrderID	Products
1	Laptop, Mouse

Fix: one row per product.

3NF Violation (Transitive Dependency):

OrderID	ZipCode	City
1	10001	New York

City depends on ZipCode, not OrderID.

Fix: move ZipCode + City to a separate ZipCodes table. Keep ZipCode on Orders as foreign key.

Normalization vs Denormalization

	Normalization	Denormalization
Goal	Reduce redundancy	Improve read performance
Joins	More joins needed	Fewer joins needed
Use case	Transactional systems (OLTP)	Reporting / analytics (OLAP)
Example	Orders → Users via foreign key	Store customer_name on orders table

Industry Examples

- **E-commerce DBs** — normalized: Products, Orders, Customers in separate tables
- **Amazon Redshift / BigQuery** — denormalized for fast analytics queries
- **Banking systems** — strictly normalized to avoid data inconsistency

Interview Questions

Question	One-Line Answer
What is normalization?	Organizing a DB to reduce redundancy by splitting data into related tables
What is 1NF?	Each column holds atomic values — no lists or repeating groups
2NF vs 3NF?	2NF removes partial key dependency; 3NF removes transitive dependency between non-key columns
What is denormalization?	Intentionally duplicating data to avoid joins — used in read-heavy or analytics systems
What problems does normalization solve?	Redundancy, update anomalies, data inconsistency